SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – SIH26214**
- **Problem Statement Title-Student Innovation**
- **Theme-Agriculture, FoodTech & Rural Development**
- **PS Category- Hardware**
- **Team ID-**
- **Team Name - RANG**

RANG

PROJECT RANG

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PROBLEM

Artisan Marginalization

Rural women have skills but lack access to high-end, modern global markets.skills but lack access to high-end, modern global markets.

The Market Gap

Conscious consumers are stuck between overpriced luxury and low-quality mass production, with no affordable ethical alternative.

Environment

Traditional cotton is water-intensive and the industry needs a sustainable alternative.

SOLUTION

Economic Empowerment (The "Who")

Transforming homemakers into entrepreneurs by upskilling them in design and financial literacy, turning "latent talent" into a steady income stream.

Affordable Sustainability (The "Why")

Bridging the market gap by offering eco-friendly, hand-painted products at competitive prices, making ethical fashion accessible to everyone.

Direct Market Linkage (The "How")

Eliminating middlemen through a Direct-to-Consumer (D2C) model, ensuring that 60-70% of the profit margin goes directly back to the women artisans.

Initial Stage

- Outsourced sustainable banana-fibre fabrics
- Low-risk, fast market entry
- Paint-to-order, low waste
- Fair & direct payments to women
- Story tag with every product

Later Stage

- Circular banana-based system
- Banana pulp will be used to make biodegradable packaging
- It is Plastic-free, low-cost packaging
- We will outsource the fabric making process and then extract pulp from the fibres to make the tags and packaging.

DIRECT CONNECT

FAIR INCOME

PAINT TO ORDER

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TECHNICAL APPROACH

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One banana stem → fabric + zero-waste packaging

BANANA PSEUDO STEM	CENTRAL CORE (10-15%)	OTHER INDUSTRIES SUCH AS CANDY,PICKLE
	BANANA FIBRE(1.5-2%)	USED FOR MAKING BANANA FABRIC FOR TOTE BAGS,BEDSHEETS ETC.
	WASTE MATERIAL (85-90%)FABRIC PULP	USED FOR MAKING THE PACKAGING

Process of making fabric and packaging from banana pseudo stem

- Banana stems are processed to provide fibres for making fabric. To crush the pseudostem we will use sugarcane crushers which are cheaper and readily available.
- The leftover bananas are pulped and processed, rather than discarding them.
- Pulp is processed into packaging material like trays and boxes.
- It becomes biodegradable and plastic packaging afterwards.

Making packaging from pulp

- The banana pseudo-stem is cut and processed to take out the fibers.
- After extraction, the fibres are washed, dried, and then spun into yarn.
- Yarn is given a weave to make banana fibre fabric.
- The residual waste pulp is then utilized in making biodegradable packaging.

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FEASIBILITY AND VIABILITY

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Feasibilty

- **Technical:** Banana pulp produces packaging 3x stronger than wood pulp with superior tear resistance.
- **Operational:** 100% circular model; leftover pulp from textile extraction directly feeds into packaging molds.
- **Economic:** Banana pulp is significantly more cost-effective to produce than traditional plastic or paper packaging.

Challenges

- •**Initial Setup Cost:** High capital required to eventually build an in-house pulp molding lab.
- **Durability Doubts:** Convincing B2B buyers that bio-packaging matches plastic's strength.
- **Transport Logistics:** Efficiently transporting heavy, wet pseudo-stems from farms to processing units.

Viability

- **Vast Raw Material:** India generates ~30M tons of pseudo-stem waste annually, providing near-free raw material.
- **Proven Model:** Similar banana-fibre businesses (e.g., Musa Pacta) process 12-13 tons/month profitably.
- **B2B Expansion:** Huge potential to sell our low-cost banana pulp packaging directly to other businesses (B2B).

Scalability

- **Integrated Craft Hubs:** Setting up dual wings: one for textiles, one dedicated to pulp-molded packaging.
- **In-House Pulp Lab:** Phasing out third-party sourcing to produce 100% of packaging in-house.
- **Bio-Plastics:** Developing banana-peel starch into compostable, transparent protective sleeves.

RISK VS SOLUTION

	Risk	Solution
1	High setup cost for Pulp Lab machinery.	Outsource pulp processing in the initial stage; reinvest early profits to build in-house capacity.
2	B2B doubts about packaging strength.	Lead marketing with certified data showing it is "3x stronger than wood pulp".
3	Inconsistent raw stem collection.	Partner with established NGO networks (like Goonj) for steady, rural supply chains.
4	Production waste management.	Zero-waste lifecycle: all stem remnants and fabric offcuts are repurposed into packaging.

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IMPACT AND BENEFITS

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Environmental Sustainability

- Banana fabric is used
- Water-based/natural dyes
- Paint-to-order so nearly zero waste
- Minimum packaging using banana pulp

Social Sustainability

- Women-led, home-based work
- Flexible hours mean increased participation.
- Skill ownership-not dependency
- Fair & transparent payments

Economic Sustainability

- Revenue-generating business model
- Repeat-use products are there so repeat income
- Profit reinvestment for scale
- No reliance on donations

Before

- Informal skill
- Middlemen dependent
- Low confidence
- Less financial knowledge
- Work seen as hobby
- No savings or financial planning

During

- Skill upgradation & quality training
- Access to raw materials & tools
- Clear product standards & timelines
- Fair pricing structure introduced

After

- Stable monthly income (₹8,500–₹9,000)
- 18–22 working days per month
- Direct payments with transparency
- Ability to price work independently

200% income growth
100% women paid directly
0 middlemen involvement
100% skill certification

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RESEARCH AND REFERENCES

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SURVEY FORM LINK:

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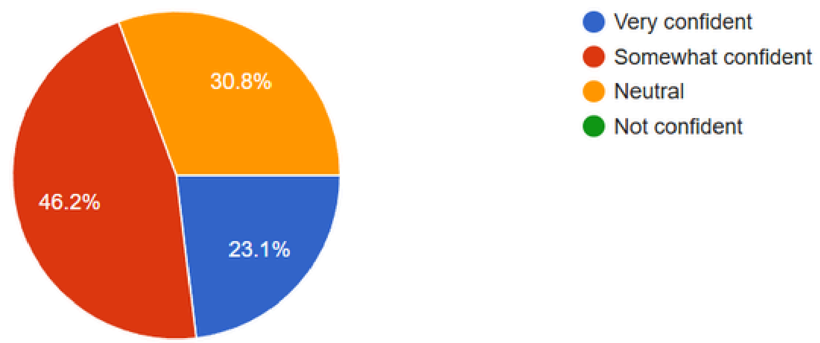
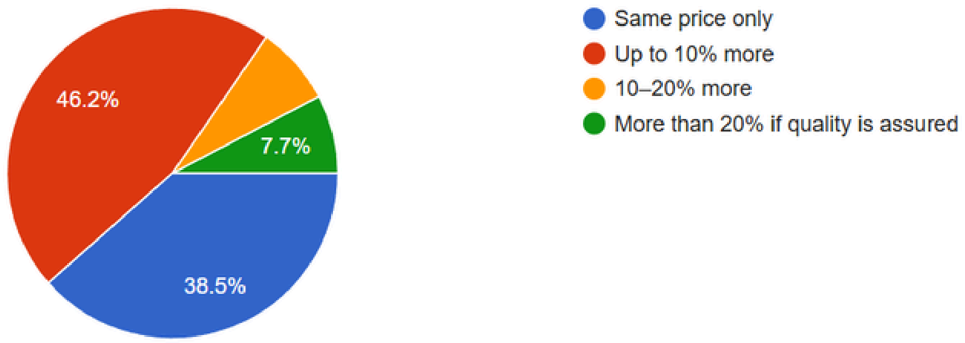
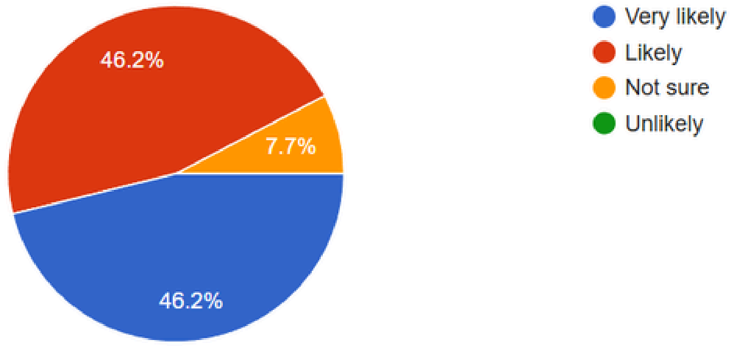
How confident would you feel using banana fibre or banana pulp products for **daily-use applications**?



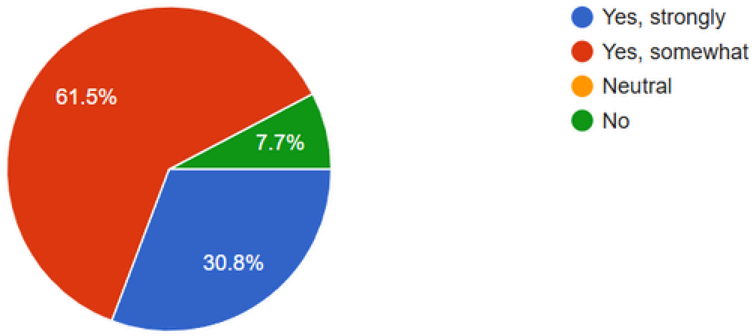
Compared to conventional products, how much extra would you be willing to pay for banana-based sustainable products?



If satisfied with quality, sustainability, and price, how likely are you to **purchase banana fibre or banana pulp products regularly**?



Does knowing that a product is made from **agricultural waste (banana fibre/pulp)** positively influence your purchase decision?



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